25. Structure constants of the Lie Algebra:

$(0,\ e^{16},\ 0,\ 0,\ 0,\ 0)$

Symplectic form

$\omega = e^{12} + e^{34} + e^{56}$

Derivatives of $3-$forms

$J, \quad d(e^{234} + (-1.0)e^{256}) = e^{1346}$

$P, \quad d(e^{234} + e^{256}) = e^{1346}$

$E, \quad d(e^{235}) = e^{1356}$

$G, \quad d(e^{245}) = e^{1456}$

$d\Lambda d$ of $3-$forms.

ODE system

ODE system with closed 3-forms

24.1. Structure constants of the Lie Algebra:

$(0,\ e^{14},\ 0,\ 0,\ e^{36},\ 0)$

Symplectic form

$\omega = e^{12} + e^{34} + e^{56}$

Derivatives of $3-$**forms**

$M, \quad d(e^{125} + (-1.0)e^{345}) = e^{1236}$

$S, \quad d(e^{125} + e^{345}) = e^{1236}$

$E, \quad d(e^{235}) = (-1.0)e^{1345}$

$C, \quad d(e^{145}) = (-1.0)e^{1346}$

$F, \quad d(e^{236}) = (-1.0)e^{1346}$

$J, \quad d(e^{234} + (-1.0)e^{256}) = (-1.0)e^{1456}$

$P, \quad d(e^{234} + e^{256}) = e^{1456}$

$G, \quad d(e^{245}) = (-1.0)e^{2346}$

$d\Lambda d$ **of** $3-$**forms.**

$E, \quad d\Lambda d(e^{235}) = e^{136} \quad B$

$G, \quad d\Lambda d(e^{245}) = (-1.0)e^{146} \quad -D$

ODE system

$\partial_t D = GDE - G^2B$

$\partial_t B = -DE^2 + GEB$

ODE system with closed 3-forms

$\partial_t D = 0$

$\partial_t B = 0$

24.2. Structure constants of the Lie Algebra:

$(0,\ e^{13},\ 0,\ 0,\ e^{46},\ 0)$

Symplectic form

$\omega = e^{12} + e^{34} + e^{56}$

Derivatives of $3-$forms

$M, \quad d(e^{125} + (-1.0)e^{345}) = e^{1246}$

$S, \quad d(e^{125} + e^{345}) = e^{1246}$

$G, \quad d(e^{245}) = e^{1345}$

$A, \quad d(e^{135}) = e^{1346}$

$H, \quad d(e^{246}) = e^{1346}$

$J, \quad d(e^{234} + (-1.0)e^{256}) = (-1.0)e^{1356}$

$P, \quad d(e^{234} + e^{256}) = e^{1356}$

$E, \quad d(e^{235}) = e^{2346}$

$d\Lambda d$ of $3-$forms.

$E, \quad d\Lambda d(e^{235}) = e^{136} \quad B$

$G, \quad d\Lambda d(e^{245}) = (-1.0)e^{146} \quad -D$

ODE system

$\partial_t D = GDE - G^2 B$

$\partial_t B = -DE^2 + GEB$

ODE system with closed 3-forms

$\partial_t D = 0$

$\partial_t B = 0$

23.1. Structure constants of the Lie Algebra:

$(0,\ (-1.0)e^{13},\ 0,\ 0,\ e^{16},\ 0)$

Symplectic form

$\omega = e^{12} + e^{34} + e^{56}$

Derivatives of $3-$forms

$E, \quad d(e^{235}) = e^{1236}$

$G, \quad d(e^{245}) = e^{1246} + (-1.0)e^{1345}$

$M, \quad d(e^{125} + (-1.0)e^{345}) = (-1.0)e^{1346}$

$H, \quad d(e^{246}) = (-1.0)e^{1346}$

$S, \quad d(e^{125} + e^{345}) = e^{1346}$

$P, \quad d(e^{234} + e^{256}) = (-1.0)e^{1356}$

$J, \quad d(e^{234} + (-1.0)e^{256}) = e^{1356}$

$d\Lambda d$ of $3-$forms.

ODE system

ODE system with closed 3-forms

23.2. Structure constants of the Lie Algebra:

$(0,\ 0,\ 0,\ e^{15},\ 0,\ e^{13})$

Symplectic form

$\omega = e^{12} + e^{34} + e^{56}$

Derivatives of $3-$forms

$H,\quad d(e^{246}) = (-1.0)e^{1234} + e^{1256}$

$J,\quad d(e^{234} + (-1.0)e^{256}) = 2.0\ e^{1235}$

$L,\quad d(e^{124} + (-1.0)e^{456}) = (-1.0)e^{1345}$

$R,\quad d(e^{124} + e^{456}) = e^{1345}$

$N,\quad d(e^{126} + (-1.0)e^{346}) = (-1.0)e^{1356}$

$T,\quad d(e^{126} + e^{346}) = e^{1356}$

$d\Lambda d$ of $3-$forms.

$H,\quad d\Lambda d(e^{246}) = (-2.0)e^{135} \quad - (2.0)A$

ODE system

$\partial_t A = (0.5)AH^2$

ODE system with closed 3-forms

$\partial_t A = 0$

22. Structure constants of the Lie Algebra:

$(0,\ e^{13},\ 0,\ 0,\ e^{16},\ 0)$

Symplectic form

$\omega = e^{12} + e^{34} + e^{56}$

Derivatives of $3-$forms

$E,\quad d(e^{235}) = e^{1236}$

$G,\quad d(e^{245}) = e^{1246} + e^{1345}$

$M,\quad d(e^{125} + (-1.0)e^{345}) = (-1.0)e^{1346}$

$S,\quad d(e^{125} + e^{345}) = e^{1346}$

$H,\quad d(e^{246}) = e^{1346}$

$J,\quad d(e^{234} + (-1.0)e^{256}) = (-1.0)e^{1356}$

$P,\quad d(e^{234} + e^{256}) = e^{1356}$

$d\Lambda d$ **of $3-$forms.**

ODE system

ODE system with closed 3-forms

21.1. Structure constants of the Lie Algebra:

$(0,\ (-1.0)e^{14}+e^{36},\ 0,\ e^{13}+e^{16},\ 0,\ 0)$

Symplectic form

$\omega = e^{12}+e^{34}+e^{56}$

Derivatives of $3-$forms

$G,\quad d(e^{245}) = e^{1235}+(-1.0)e^{1256}+e^{3456}$

$H,\quad d(e^{246}) = e^{1236}$

$P,\quad d(e^{234}+e^{256}) = e^{1236}+(-1.0)e^{1456}$

$J,\quad d(e^{234}+(-1.0)e^{256}) = e^{1236}+e^{1456}$

$E,\quad d(e^{235}) = e^{1345}$

$F,\quad d(e^{236}) = e^{1346}$

$L,\quad d(e^{124}+(-1.0)e^{456}) = e^{1346}+(-1.0)e^{1356}$

$R,\quad d(e^{124}+e^{456}) = e^{1346}+e^{1356}$

$M,\quad d(e^{125}+(-1.0)e^{345}) = 2.0\ e^{1356}$

$d\Lambda d$ of $3-$forms.

$G,\quad d\Lambda d(e^{245}) = 2.0\ e^{136}\quad (2.0)B$

ODE system

$\partial_t B = (0.5)G^2 B$

ODE system with closed 3-forms

$\partial_t B = 0$

21.2. Structure constants of the Lie Algebra:

$(0,\ e^{14}+e^{36},\ 0,\ e^{16},\ 0,\ 0)$

Symplectic form

$\omega = e^{12}+e^{34}+e^{56}$

Derivatives of $3-$forms

$J,\quad d(e^{234}+(-1.0)e^{256}) = e^{1236}+(-1.0)e^{1456}$

$P,\quad d(e^{234}+e^{256}) = e^{1236}+e^{1456}$

$G,\quad d(e^{245}) = (-1.0)e^{1256}+e^{3456}$

$E,\quad d(e^{235}) = (-1.0)e^{1345}$

$F,\quad d(e^{236}) = (-1.0)e^{1346}$

$L,\quad d(e^{124}+(-1.0)e^{456}) = e^{1346}$

$R,\quad d(e^{124}+e^{456}) = e^{1346}$

$M,\quad d(e^{125}+(-1.0)e^{345}) = 2.0\ e^{1356}$

$d\Lambda d$ of $3-$forms.

$G,\quad d\Lambda d(e^{245}) = 2.0\ e^{136}\quad (2.0)B$

ODE system

$\partial_t B = (0.5)G^2 B$

ODE system with closed 3-forms

$\partial_t B = 0$

20. Structure constants of the Lie Algebra:

$(0,\ 0,\ e^{16},\ (-1.0)e^{13},\ e^{14} + (-1.0)e^{36},\ 0)$

Symplectic form

$\omega = e^{12} + e^{34} + e^{56}$

Derivatives of 3−forms

$E,\quad d(e^{235}) = e^{1234} + (-1.0)e^{1256}$

$G,\quad d(e^{245}) = (-1.0)e^{1235} + e^{2346}$

$H,\quad d(e^{246}) = (-1.0)e^{1236}$

$M,\quad d(e^{125} + (-1.0)e^{345}) = (-1.0)e^{1236} + (-1.0)e^{1456}$

$S,\quad d(e^{125} + e^{345}) = (-1.0)e^{1236} + e^{1456}$

$J,\quad d(e^{234} + (-1.0)e^{256}) = (-2.0)e^{1246}$

$K,\quad d(e^{123} + (-1.0)e^{356}) = (-1.0)e^{1346}$

$Q,\quad d(e^{123} + e^{356}) = e^{1346}$

$C,\quad d(e^{145}) = e^{1346}$

$R,\quad d(e^{124} + e^{456}) = (-1.0)e^{1356}$

$L,\quad d(e^{124} + (-1.0)e^{456}) = e^{1356}$

$d\Lambda d$ of 3−forms.

$G,\quad d\Lambda d(e^{245}) = (-1.0)e^{134} + e^{156} \quad - I$

$J,\quad d\Lambda d(e^{234} + (-1.0)e^{256}) = 2.0\ e^{136} \quad (2.0)B$

$E,\quad d\Lambda d(e^{235}) = (-2.0)e^{146} \quad - (2.0)D$

ODE system

$\partial_t D = (0.5)DE^2 + JEI - (0.5)GEB$

$\partial_t B = (HE - GF - P^2)I + (0.5)JGB - (0.5)JDE$

$\partial_t I = (2.0)JGI + GDE - G^2B$

ODE system with closed 3-forms

$\partial_t D = 0$

$\partial_t B = -P^2I$

$\partial_t I = 0$

19. Structure constants of the Lie Algebra:

$(0,\ 0,\ e^{14},\ e^{15},\ 0,\ e^{13})$

Symplectic form

$\omega = e^{12} + e^{34} + e^{56}$

Derivatives of $3-$forms

$H,\quad d(e^{246}) = (-1.0)e^{1234} + e^{1256}$

$J,\quad d(e^{234} + (-1.0)e^{256}) = 2.0\ e^{1235}$

$E,\quad d(e^{235}) = e^{1245}$

$F,\quad d(e^{236}) = e^{1246}$

$L,\quad d(e^{124} + (-1.0)e^{456}) = (-1.0)e^{1345}$

$R,\quad d(e^{124} + e^{456}) = e^{1345}$

$N,\quad d(e^{126} + (-1.0)e^{346}) = (-1.0)e^{1356}$

$T,\quad d(e^{126} + e^{346}) = e^{1356}$

$K,\quad d(e^{123} + (-1.0)e^{356}) = (-1.0)e^{1456}$

$Q,\quad d(e^{123} + e^{356}) = e^{1456}$

$d\Lambda d$ of $3-$forms.

$F,\quad d\Lambda d(e^{236}) = (-1.0)e^{134} + e^{156}\quad - I$

$H,\quad d\Lambda d(e^{246}) = (-2.0)e^{135}\quad - (2.0)A$

$J,\quad d\Lambda d(e^{234} + (-1.0)e^{256}) = 2.0\ e^{145}\quad (2.0)C$

ODE system

$\partial_t C = -(0.5)JAH + (0.5)CJF + (HE - GF - P^2)I$

$\partial_t A = JHI + (0.5)AH^2 - (0.5)CHF$

$\partial_t I = AHF - CF^2 + (2.0)JIF$

ODE system with closed 3-forms

$\partial_t C = -P^2 I$

$\partial_t A = 0$

$\partial_t I = 0$

18.1 Structure constants of the Lie Algebra:

$(0,\ (-2.0)e^{36},\ 0,\ (-1.0)e^{16},\ (-1.0)e^{13},\ 0)$

Symplectic form

$\omega = e^{12} + e^{34} + e^{56}$

Derivatives of $3-$forms

$G,\quad d(e^{245}) = e^{1234} + e^{1256} + (-2.0)e^{3456}$

$P,\quad d(e^{234} + e^{256}) = (-2.0)e^{1236}$

$L,\quad d(e^{124} + (-1.0)e^{456}) = (-3.0)e^{1346}$

$R,\quad d(e^{124} + e^{456}) = (-1.0)e^{1346}$

$M,\quad d(e^{125} + (-1.0)e^{345}) = (-3.0)e^{1356}$

$S,\quad d(e^{125} + e^{345}) = (-1.0)e^{1356}$

$d\Lambda d$ of $3-$forms.

$G,\quad d\Lambda d(e^{245}) = 6.0\ e^{136}\quad (6.0)B$

ODE system

$\partial_t B = (0.16666666666666666667)G^2 B$

ODE system with closed 3-forms

$\partial_t B = 0$

18.2. Structure constants of the Lie Algebra:

$(0,\ (-1.0)e^{13}+(-1.0)e^{35},\ 0,\ e^{15},\ 0,\ 0.5\ e^{35})$

Symplectic form

$\omega = e^{12}+e^{34}+e^{56}$

Derivatives of 3−forms

$N,\quad d(e^{126}+(-1.0)e^{346}) = 0.5\ e^{1235}$

$T,\quad d(e^{126}+e^{346}) = 0.5\ e^{1235}+2.0\ e^{1356}$

$P,\quad d(e^{234}+e^{256}) = e^{1235}+(-1.0)e^{1356}$

$J,\quad d(e^{234}+(-1.0)e^{256}) = e^{1235}+e^{1356}$

$H,\quad d(e^{246}) = e^{1256}+(-1.0)e^{1346}+(-0.5)e^{2345}+e^{3456}$

$L,\quad d(e^{124}+(-1.0)e^{456}) = (-1.0)e^{1345}$

$R,\quad d(e^{124}+e^{456}) = (-1.0)e^{1345}$

$G,\quad d(e^{245}) = (-1.0)e^{1345}$

$D,\quad d(e^{146}) = (-0.5)e^{1345}$

$d\Lambda d$ of 3−forms.

$H,\quad d\Lambda d(e^{246}) = 3.0\ e^{135}\quad (3.0)A$

ODE system

$\partial_t A = -(0.33333333333333333334)AH^2$

ODE system with closed 3-forms

$\partial_t A = 0$

18.3. Structure constants of the Lie Algebra:

$(0,\ e^{36},\ 0,\ e^{13} + 2.0\ e^{16},\ (-1.0)e^{13},\ 0)$

Symplectic form

$\omega = e^{12} + e^{34} + e^{56}$

Derivatives of $3-$forms

$G, \quad d(e^{245}) = e^{1234} + e^{1235} + (-2.0)e^{1256} + e^{3456}$

$P, \quad d(e^{234} + e^{256}) = e^{1236}$

$H, \quad d(e^{246}) = e^{1236}$

$J, \quad d(e^{234} + (-1.0)e^{256}) = 3.0\ e^{1236}$

$R, \quad d(e^{124} + e^{456}) = 2.0\ e^{1346} + e^{1356}$

$L, \quad d(e^{124} + (-1.0)e^{456}) = (-1.0)e^{1356}$

$S, \quad d(e^{125} + e^{345}) = (-1.0)e^{1356}$

$M, \quad d(e^{125} + (-1.0)e^{345}) = 3.0\ e^{1356}$

$d\Lambda d$ of $3-$forms.

$G, \quad d\Lambda d(e^{245}) = 6.0\ e^{136} \quad (6.0)B$

ODE system

$\partial_t B = (0.16666666666666666667)G^2 B$

ODE system with closed 3-forms

$\partial_t B = 0$

17. Structure constants of the Lie Algebra:

$(0,\ (-1.0)e^{14} + e^{36},\ 0,\ 0,\ e^{13},\ 0)$

Symplectic form

$\omega = e^{12} + e^{34} + e^{56}$

Derivatives of $3-$forms

$G,\quad d(e^{245}) = (-1.0)e^{1234} + e^{3456}$

$J,\quad d(e^{234} + (-1.0)e^{256}) = (-1.0)e^{1236} + e^{1456}$

$P,\quad d(e^{234} + e^{256}) = e^{1236} + (-1.0)e^{1456}$

$E,\quad d(e^{235}) = e^{1345}$

$F,\quad d(e^{236}) = e^{1346}$

$L,\quad d(e^{124} + (-1.0)e^{456}) = 2.0\ e^{1346}$

$M,\quad d(e^{125} + (-1.0)e^{345}) = e^{1356}$

$S,\quad d(e^{125} + e^{345}) = e^{1356}$

$d\Lambda d$ of $3-$forms.

$G,\quad d\Lambda d(e^{245}) = 2.0\ e^{136}\quad (2.0)B$

ODE system

$\partial_t B = (0.5)G^2 B$

ODE system with closed 3-forms

$\partial_t B = 0$

16.1. Structure constants of the Lie Algebra:

$(0,\ (-1.0)e^{14} + (-1.0)e^{36},\ 0,\ 0,\ (-1.0)e^{13} + e^{46},\ 0)$

Symplectic form

$\omega = e^{12} + e^{34} + e^{56}$

Derivatives of $3-$forms

$G,\quad d(e^{245}) = e^{1234} + (-1.0)e^{3456}$

$P,\quad d(e^{234} + e^{256}) = (-1.0)e^{1236} + (-1.0)e^{1456}$

$J,\quad d(e^{234} + (-1.0)e^{256}) = e^{1236} + e^{1456}$

$M,\quad d(e^{125} + (-1.0)e^{345}) = e^{1246} + (-1.0)e^{1356}$

$S,\quad d(e^{125} + e^{345}) = e^{1246} + (-1.0)e^{1356}$

$E,\quad d(e^{235}) = e^{1345} + e^{2346}$

$L,\quad d(e^{124} + (-1.0)e^{456}) = (-2.0)e^{1346}$

$A,\quad d(e^{135}) = e^{1346}$

$F,\quad d(e^{236}) = e^{1346}$

$d\Lambda d$ of $3-$forms.

$G,\quad d\Lambda d(e^{245}) = 2.0\ e^{136}\quad (2.0)B$

$E,\quad d\Lambda d(e^{235}) = (-2.0)e^{146}\quad -(2.0)D$

ODE system

$\partial_t D = (0.5)DE^2 - (0.5)GEB$

$\partial_t B = -(0.5)GDE + (0.5)G^2B$

ODE system with closed 3-forms

$\partial_t D = 0$

$\partial_t B = 0$

16.2. Structure constants of the Lie Algebra:

$(0,\ (-1.0)e^{13},\ (-1.0)e^{45},\ 0,\ 0,\ 0)$

Symplectic form

$\omega = e^{12} + e^{34} + e^{56}$

Derivatives of $3-$forms

$K,\quad d(e^{123} + (-1.0)e^{356}) = (-1.0)e^{1245}$

$Q,\quad d(e^{123} + e^{356}) = (-1.0)e^{1245}$

$G,\quad d(e^{245}) = (-1.0)e^{1345}$

$H,\quad d(e^{246}) = (-1.0)e^{1346}$

$P,\quad d(e^{234} + e^{256}) = (-1.0)e^{1356}$

$J,\quad d(e^{234} + (-1.0)e^{256}) = e^{1356}$

$B,\quad d(e^{136}) = e^{1456}$

$F,\quad d(e^{236}) = e^{2456}$

$d\Lambda d$ of $3-$forms.

$F,\quad d\Lambda d(e^{236}) = (-1.0)e^{134} \quad - (0.5)O - (0.5)I$

$P,\quad d\Lambda d(e^{234} + e^{256}) = (-1.0)e^{145} \quad - C$

$J,\quad d\Lambda d(e^{234} + (-1.0)e^{256}) = e^{145} \quad C$

ODE system

ODE system with closed 3-forms

15.1. Structure constants of the Lie Algebra:

$(0,\ (-1.0)e^{14},\ 0,\ e^{16},\ (-1.0)e^{13}+(-1.0)e^{36},\ 0)$

Symplectic form

$\omega = e^{12}+e^{34}+e^{56}$

Derivatives of $3-$**forms**

$G,\quad d(e^{245}) = e^{1234}+(-1.0)e^{1256}+e^{2346}$

$S,\quad d(e^{125}+e^{345}) = (-1.0)e^{1236}+(-1.0)e^{1356}$

$M,\quad d(e^{125}+(-1.0)e^{345}) = (-1.0)e^{1236}+e^{1356}$

$J,\quad d(e^{234}+(-1.0)e^{256}) = 2.0\ e^{1236}+e^{1456}$

$E,\quad d(e^{235}) = e^{1345}$

$L,\quad d(e^{124}+(-1.0)e^{456}) = (-1.0)e^{1346}$

$R,\quad d(e^{124}+e^{456}) = e^{1346}$

$C,\quad d(e^{145}) = e^{1346}$

$F,\quad d(e^{236}) = e^{1346}$

$P,\quad d(e^{234}+e^{256}) = (-1.0)e^{1456}$

$d\Lambda d$ **of** $3-$**forms.**

$E,\quad d\Lambda d(e^{235}) = e^{136}\quad B$

$G,\quad d\Lambda d(e^{245}) = 2.0\ e^{136}+(-1.0)e^{146}\quad -D+(2.0)B$

ODE system

$\partial_t D = (GE-(2.0)E^2)D+(-G^2+(2.0)GE)B$

$\partial_t B = -DE^2+GEB$

ODE system with closed 3-forms

$\partial_t D = 0$

$\partial_t B = 0$

15.2. Structure constants of the Lie Algebra:

$((-1.0)e^{23} + e^{36},\ 0,\ e^{26},\ 0,\ e^{36},\ 0)$

Symplectic form

$\omega = e^{12} + e^{34} + e^{56}$

Derivatives of $3-$forms

$M, \quad d(e^{125} + (-1.0)e^{345}) = e^{1236} + (-1.0)e^{2356} + (-1.0)e^{2456}$

$S, \quad d(e^{125} + e^{345}) = e^{1236} + (-1.0)e^{2356} + e^{2456}$

$O, \quad d(e^{134} + e^{156}) = e^{1246} + (-1.0)e^{2356}$

$I, \quad d(e^{134} + (-1.0)e^{156}) = e^{1246} + e^{2356}$

$A, \quad d(e^{135}) = e^{1256}$

$C, \quad d(e^{145}) = (-1.0)e^{1346} + (-1.0)e^{2345} + e^{3456}$

$L, \quad d(e^{124} + (-1.0)e^{456}) = (-1.0)e^{2346}$

$R, \quad d(e^{124} + e^{456}) = (-1.0)e^{2346}$

$D, \quad d(e^{146}) = (-1.0)e^{2346}$

$G, \quad d(e^{245}) = (-1.0)e^{2346}$

$d\Lambda d$ of $3-$forms.

$A, \quad d\Lambda d(e^{135}) = e^{236} \quad F$

$C, \quad d\Lambda d(e^{145}) = 2.0\ e^{236} + (-1.0)e^{246} \quad -H + (2.0)F$

ODE system

$\partial_t H = (C^2 - (2.0)CA)F + ((2.0)A^2 - CA)H$

$\partial_t F = A^2 H - CAF$

ODE system with closed 3-forms

$\partial_t H = 0$

$\partial_t F = 0$

15.3. Structure constants of the Lie Algebra:

$(0,\ (-1.0)e^{14},\ 0,\ (-1.0)e^{13},\ 0,\ (-1.0)e^{35})$

Symplectic form

$\omega = e^{12} + e^{34} + e^{56}$

Derivatives of $3-$**forms**

$N,\quad d(e^{126} + (-1.0)e^{346}) = (-1.0)e^{1235}$

$T,\quad d(e^{126} + e^{346}) = (-1.0)e^{1235}$

$G,\quad d(e^{245}) = (-1.0)e^{1235}$

$H,\quad d(e^{246}) = (-1.0)e^{1236} + e^{2345}$

$D,\quad d(e^{146}) = e^{1345}$

$E,\quad d(e^{235}) = e^{1345}$

$F,\quad d(e^{236}) = e^{1346}$

$R,\quad d(e^{124} + e^{456}) = (-1.0)e^{1356}$

$L,\quad d(e^{124} + (-1.0)e^{456}) = e^{1356}$

$P,\quad d(e^{234} + e^{256}) = (-1.0)e^{1456}$

$J,\quad d(e^{234} + (-1.0)e^{256}) = e^{1456}$

$d\Lambda d$ **of** $3-$**forms.**

$F,\quad d\Lambda d(e^{236}) = e^{135}\quad A$

$H,\quad d\Lambda d(e^{246}) = (-1.0)e^{145}\quad -C$

ODE system

$\partial_t C = AH^2 - CHF$

$\partial_t A = -AHF + CF^2$

ODE system with closed 3-forms

$\partial_t C = 0$

$\partial_t A = 0$

14.1. Structure constants of the Lie Algebra:

$(0,\ (-1.0)e^{14},\ 0,\ (-1.0)e^{13},\ 0,\ (-1.0)e^{15})$

Symplectic form

$\omega = e^{12} + e^{34} + e^{56}$

Derivatives of $3-$forms

$G,\quad d(e^{245}) = (-1.0)e^{1235}$

$F,\quad d(e^{236}) = (-1.0)e^{1235} + e^{1346}$

$H,\quad d(e^{246}) = (-1.0)e^{1236} + (-1.0)e^{1245}$

$T,\quad d(e^{126} + e^{346}) = (-1.0)e^{1345}$

$N,\quad d(e^{126} + (-1.0)e^{346}) = e^{1345}$

$E,\quad d(e^{235}) = e^{1345}$

$R,\quad d(e^{124} + e^{456}) = (-1.0)e^{1356}$

$L,\quad d(e^{124} + (-1.0)e^{456}) = e^{1356}$

$P,\quad d(e^{234} + e^{256}) = (-1.0)e^{1456}$

$J,\quad d(e^{234} + (-1.0)e^{256}) = e^{1456}$

$d\Lambda d$ of $3-$forms.

$H,\quad d\Lambda d(e^{246}) = 2.0\ e^{135}\quad (2.0)A$

ODE system

$\partial_t A = -(0.5)AH^2$

ODE system with closed 3-forms

$\partial_t A = 0$

14.2. Structure constants of the Lie Algebra:

$(e^{25},\ 0,\ 0,\ e^{35},\ 0,\ e^{15})$

Symplectic form

$\omega = e^{12} + e^{34} + e^{56}$

Derivatives of $3-$forms

$L, \quad d(e^{124} + (-1.0)e^{456}) = e^{1235}$

$R, \quad d(e^{124} + e^{456}) = e^{1235}$

$F, \quad d(e^{236}) = e^{1235}$

$H, \quad d(e^{246}) = e^{1245} + (-1.0)e^{2356}$

$N, \quad d(e^{126} + (-1.0)e^{346}) = (-1.0)e^{1345}$

$T, \quad d(e^{126} + e^{346}) = e^{1345}$

$D, \quad d(e^{146}) = (-1.0)e^{1356} + (-1.0)e^{2456}$

$I, \quad d(e^{134} + (-1.0)e^{156}) = e^{2345}$

$O, \quad d(e^{134} + e^{156}) = e^{2345}$

$B, \quad d(e^{136}) = (-1.0)e^{2356}$

$d\Lambda d$ of $3-$forms.

$D, \quad d\Lambda d(e^{146}) = 2.0\ e^{235} \quad (2.0)E$

ODE system

$\partial_t E = (0.5)D^2 E$

ODE system with closed 3-forms

$\partial_t E = 0$

14.3 Structure constants of the Lie Algebra:

$(0,\ (-1.0)e^{14},\ 0,\ (-1.0)e^{15},\ 0,\ (-1.0)e^{13})$

Symplectic form

$\omega = e^{12} + e^{34} + e^{56}$

Derivatives of 3−forms

$H,\quad d(e^{246}) = e^{1234} + (-1.0)e^{1256}$

$J,\quad d(e^{234} + (-1.0)e^{256}) = (-2.0)e^{1235} + e^{1456}$

$R,\quad d(e^{124} + e^{456}) = (-1.0)e^{1345}$

$L,\quad d(e^{124} + (-1.0)e^{456}) = e^{1345}$

$E,\quad d(e^{235}) = e^{1345}$

$F,\quad d(e^{236}) = e^{1346}$

$T,\quad d(e^{126} + e^{346}) = (-1.0)e^{1356}$

$N,\quad d(e^{126} + (-1.0)e^{346}) = e^{1356}$

$P,\quad d(e^{234} + e^{256}) = (-1.0)e^{1456}$

$d\Lambda d$ of 3−forms.

$H,\quad d\Lambda d(e^{246}) = (-2.0)e^{135} \quad - (2.0)A$

ODE system

$\partial_t A = (0.5)AH^2$

ODE system with closed 3-forms

$\partial_t A = 0$

10. Structure constants of the Lie Algebra:

$(0,\ (-1.0)e^{14}+e^{35}+e^{45},\ (-1.0)e^{15},\ 0,\ 0,\ (-1.0)e^{13}+e^{35})$

Symplectic form

$\omega = e^{12}+e^{34}+e^{56}$

Derivatives of $3-$**forms**

$H,\quad d(e^{246}) = e^{1234}+(-1.0)e^{2345}+(-1.0)e^{3456}$

$J,\quad d(e^{234}+(-1.0)e^{256}) = (-1.0)e^{1235}+e^{1245}+e^{1456}$

$P,\quad d(e^{234}+e^{256}) = e^{1235}+e^{1245}+(-1.0)e^{1456}$

$T,\quad d(e^{126}+e^{346}) = e^{1235}+(-1.0)e^{1356}$

$N,\quad d(e^{126}+(-1.0)e^{346}) = e^{1235}+(-1.0)e^{1356}+(-2.0)e^{1456}$

$F,\quad d(e^{236}) = (-1.0)e^{1256}+e^{1346}+e^{3456}$

$K,\quad d(e^{123}+(-1.0)e^{356}) = (-1.0)e^{1345}$

$Q,\quad d(e^{123}+e^{356}) = (-1.0)e^{1345}$

$D,\quad d(e^{146}) = (-1.0)e^{1345}$

$E,\quad d(e^{235}) = e^{1345}$

$L,\quad d(e^{124}+(-1.0)e^{456}) = 2.0\ e^{1345}$

$d\Lambda d$ **of** $3-$**forms.**

$H,\quad d\Lambda d(e^{246}) = (-2.0)e^{135}\quad -(2.0)A$

$F,\quad d\Lambda d(e^{236}) = 2.0\ e^{145}\quad (2.0)C$

ODE system

$\partial_t C = -(0.5)AHF+(0.5)CF^2$

$\partial_t A = (0.5)AH^2-(0.5)CHF$

ODE system with closed 3-forms

$\partial_t C = 0$

$\partial_t A = 0$

6.1. Structure constants of the Lie Algebra:

$(0,\ e^{35}+e^{45},\ e^{45},\ e^{15},\ 0,\ (-1.0)e^{14})$

Symplectic form

$\omega = e^{12}+e^{34}+e^{56}$

Derivatives of 3−forms

$F,\quad d(e^{236}) = (-1.0)e^{1234} + (-1.0)e^{2456} + e^{3456}$

$J,\quad d(e^{234} + (-1.0)e^{256}) = e^{1235} + (-1.0)e^{1245}$

$P,\quad d(e^{234} + e^{256}) = e^{1235} + e^{1245}$

$Q,\quad d(e^{123} + e^{356}) = e^{1245}$

$K,\quad d(e^{123} + (-1.0)e^{356}) = e^{1245} + (-2.0)e^{1345}$

$H,\quad d(e^{246}) = e^{1256} + (-1.0)e^{3456}$

$L,\quad d(e^{124} + (-1.0)e^{456}) = e^{1345}$

$R,\quad d(e^{124} + e^{456}) = e^{1345}$

$N,\quad d(e^{126} + (-1.0)e^{346}) = (-2.0)e^{1356} + (-1.0)e^{1456}$

$T,\quad d(e^{126} + e^{346}) = (-1.0)e^{1456}$

$B,\quad d(e^{136}) = (-1.0)e^{1456}$

$d\Lambda d$ of 3−forms.

$F,\quad d\Lambda d(e^{236}) = (-1.0)e^{125} + e^{135} + 2.0\ e^{145} + e^{345}\quad (2.0)C - M + A$

$H,\quad d\Lambda d(e^{246}) = (-2.0)e^{135} + (-1.0)e^{145}\quad - C - (2.0)A$

$N,\quad d\Lambda d(e^{126} + (-1.0)e^{346}) = 2.0\ e^{145}\quad (2.0)C$

ODE system

$\partial_t C = M(HB - T^2 - DF) - (0.5)ANH + (0.5)CNF$

$\partial_t M = (-(0.75)NH + HF + (0.5)H^2)A + M((1.5)HB - (1.5)T^2 + NH - (1.5)DF + (2.0)NF) + C(-F^2 - (0.5)HF + (0.75)NF)$

$\partial_t A = A((0.25)NH + (0.5)H^2) + M(-(0.5)HB + (0.5)T^2 + NH + (0.5)DF) + C(-(0.5)HF - (0.25)NF)$

ODE system with closed 3-forms

$\partial_t C = 0$

$\partial_t M = 0$

$\partial_t A = 0$

6.2. Structure constants of the Lie Algebra:

$(e^{36} + e^{46},\ 0,\ e^{26},\ e^{36},\ (-1.0)e^{23},\ 0)$

Symplectic form

$\omega = e^{12} + e^{34} + e^{56}$

Derivatives of 3−forms

$C,\quad d(e^{145}) = (-1.0)e^{1234} + e^{1356} + e^{3456}$

$I,\quad d(e^{134} + (-1.0)e^{156}) = (-1.0)e^{1236} + e^{1246}$

$R,\quad d(e^{124} + e^{456}) = e^{1236}$

$O,\quad d(e^{134} + e^{156}) = e^{1236} + e^{1246}$

$L,\quad d(e^{124} + (-1.0)e^{456}) = e^{1236} + (-2.0)e^{2346}$

$A,\quad d(e^{135}) = e^{1256} + (-1.0)e^{3456}$

$K,\quad d(e^{123} + (-1.0)e^{356}) = e^{2346}$

$Q,\quad d(e^{123} + e^{356}) = e^{2346}$

$S,\quad d(e^{125} + e^{345}) = (-1.0)e^{2356}$

$M,\quad d(e^{125} + (-1.0)e^{345}) = (-1.0)e^{2356} + (-2.0)e^{2456}$

$G,\quad d(e^{245}) = e^{2356}$

$d\Lambda d$ of 3−forms.

$C,\quad d\Lambda d(e^{145}) = (-1.0)e^{126} + (-2.0)e^{236} + (-1.0)e^{246} + e^{346} \quad -N - H - (2.0)F$

$A,\quad d\Lambda d(e^{135}) = e^{236} + 2.0\ e^{246} \quad (2.0)H + F$

$M,\quad d\Lambda d(e^{125} + (-1.0)e^{345}) = 2.0\ e^{236} \quad (2.0)F$

ODE system

$\partial_t N = (-(1.5)CE - (2.0)CM - MA - (1.5)S^2 + (1.5)GA)N + (-(0.5)A^2 - (0.75)MA - CA)H + ((0.75)CM + C^2 + (0.5)CA)F$

$\partial_t H = N(-(0.5)CE + MA - (0.5)S^2 + (0.5)GA) + ((0.25)CM - (0.5)CA)F + H((0.5)A^2 - (0.25)MA)$

$\partial_t F = -(0.5)CMF + (0.5)MAH + (CE + S^2 - GA)N$

ODE system with closed 3-forms

$\partial_t N = 0$

$\partial_t H = 0$

$\partial_t F = 0$

3. Structure constants of the Lie Algebra:

$(0,\ (-1.0)e^{15},\ (-1.0)e^{16},\ (-1.0)e^{13},\ (-1.0)e^{14},\ 0)$

Symplectic form

$\omega = e^{12} + e^{34} + e^{56}$

Derivatives of $3-$forms

$E, \quad d(e^{235}) = (-1.0)e^{1234} + e^{1256}$

$G, \quad d(e^{245}) = (-1.0)e^{1235}$

$H, \quad d(e^{246}) = (-1.0)e^{1236} + e^{1456}$

$J, \quad d(e^{234} + (-1.0)e^{256}) = 2.0\ e^{1246} + (-1.0)e^{1345}$

$P, \quad d(e^{234} + e^{256}) = (-1.0)e^{1345}$

$Q, \quad d(e^{123} + e^{356}) = (-1.0)e^{1346}$

$K, \quad d(e^{123} + (-1.0)e^{356}) = e^{1346}$

$R, \quad d(e^{124} + e^{456}) = (-1.0)e^{1356}$

$L, \quad d(e^{124} + (-1.0)e^{456}) = e^{1356}$

$F, \quad d(e^{236}) = e^{1356}$

$S, \quad d(e^{125} + e^{345}) = (-1.0)e^{1456}$

$M, \quad d(e^{125} + (-1.0)e^{345}) = e^{1456}$

$d\Lambda d$ of $3-$forms.

$G, \quad d\Lambda d(e^{245}) = e^{134} + (-1.0)e^{156} \quad I$

$J, \quad d\Lambda d(e^{234} + (-1.0)e^{256}) = (-2.0)e^{136} \quad - (2.0)B$

$E, \quad d\Lambda d(e^{235}) = (-2.0)e^{146} \quad - (2.0)D$

ODE system

$\partial_t D = (0.5)DE^2 + JEI - (0.5)GEB$

$\partial_t B = (-HE + GF + P^2)I - (0.5)JGB + (0.5)JDE$

$\partial_t I = -2JGI - GDE + G^2B$

ODE system with closed 3-forms

$\partial_t D = 0$

$\partial_t B = 0$

$\partial_t I = 0$

2. Structure constants of the Lie Algebra:

$(0,\ (-1.0)e^{14} + (-1.0)e^{15} + e^{36},\ (-1.0)e^{16},\ (-1.0)e^{13},\ e^{13} + (-1.0)e^{14},\ 0)$

Symplectic form

$\omega = e^{12} + e^{34} + e^{56}$

Derivatives of 3−forms

$G,\quad d(e^{245}) = (-1.0)e^{1234} + (-1.0)e^{1235} + e^{3456}$

$E,\quad d(e^{235}) = (-1.0)e^{1234} + e^{1256} + e^{1345}$

$J,\quad d(e^{234} + (-1.0)e^{256}) = (-1.0)e^{1236} + 2.0\ e^{1246} + (-1.0)e^{1345} + e^{1456}$

$H,\quad d(e^{246}) = (-1.0)e^{1236} + e^{1456}$

$P,\quad d(e^{234} + e^{256}) = e^{1236} + (-1.0)e^{1345} + (-1.0)e^{1456}$

$Q,\quad d(e^{123} + e^{356}) = (-1.0)e^{1346}$

$K,\quad d(e^{123} + (-1.0)e^{356}) = e^{1346}$

$F,\quad d(e^{236}) = e^{1346} + e^{1356}$

$L,\quad d(e^{124} + (-1.0)e^{456}) = 2.0\ e^{1346} + e^{1356}$

$R,\quad d(e^{124} + e^{456}) = (-1.0)e^{1356}$

$S,\quad d(e^{125} + e^{345}) = e^{1356} + (-1.0)e^{1456}$

$M,\quad d(e^{125} + (-1.0)e^{345}) = e^{1356} + e^{1456}$

$d\Lambda d$ of 3−forms.

$G,\quad d\Lambda d(e^{245}) = e^{134} + 2.0\ e^{136} + (-1.0)e^{146} + (-1.0)e^{156} \quad - D + (2.0)B + I$

$J,\quad d\Lambda d(e^{234} + (-1.0)e^{256}) = (-2.0)e^{136} \quad - (2.0)B$

$E,\quad d\Lambda d(e^{235}) = e^{136} + (-2.0)e^{146} \quad - (2.0)D + B$

ODE system

$\partial_t D = ((0.25)JE + (0.5)E^2)D + (-(0.5)HE + JE + (0.5)GF + (0.5)P^2)I + (-(0.5)GE - (0.25)JG)B$

$\partial_t B = (-HE + GF + P^2)I - (0.5)JGB + (0.5)JDE$

$\partial_t I = (G^2 - (0.5)GE + (0.75)JG)B - D(GE + (0.75)JE - (0.5)E^2) + ((1.5)HE - 2JG + JE - (1.5)GF - (1.5)P^2)I$

ODE system with closed 3-forms

$\partial_t D = 0$

$\partial_t B = 0$

$\partial_t I = 0$

0. Structure constants of the Lie Algebra:

$(0,\ 0,\ (-1.0)e^{13},\ e^{14},\ (-1.0)e^{25},\ e^{26})$

Symplectic form

$\omega = e^{12} + e^{34} + e^{56}$

Derivatives of 3−forms

$E,\quad d(e^{235}) = (-1.0)e^{1235}$

$A,\quad d(e^{135}) = e^{1235}$

$B,\quad d(e^{136}) = (-1.0)e^{1236}$

$F,\quad d(e^{236}) = (-1.0)e^{1236}$

$C,\quad d(e^{145}) = e^{1245}$

$G,\quad d(e^{245}) = e^{1245}$

$D,\quad d(e^{146}) = (-1.0)e^{1246}$

$H,\quad d(e^{246}) = e^{1246}$

$Q,\quad d(e^{123} + e^{356}) = (-1.0)e^{1356}$

$K,\quad d(e^{123} + (-1.0)e^{356}) = e^{1356}$

$L,\quad d(e^{124} + (-1.0)e^{456}) = (-1.0)e^{1456}$

$R,\quad d(e^{124} + e^{456}) = e^{1456}$

$S,\quad d(e^{125} + e^{345}) = (-1.0)e^{2345}$

$M,\quad d(e^{125} + (-1.0)e^{345}) = e^{2345}$

$N,\quad d(e^{126} + (-1.0)e^{346}) = (-1.0)e^{2346}$

$T,\quad d(e^{126} + e^{346}) = e^{2346}$

$d\Lambda d$ of 3−forms.

$A,\quad d\Lambda d(e^{135}) = (-1.0)e^{135} + (-1.0)e^{235} \quad -A - E$

$E,\quad d\Lambda d(e^{235}) = e^{135} + e^{235} \quad A + E$

$B,\quad d\Lambda d(e^{136}) = e^{136} + (-1.0)e^{236} \quad B - F$

$F,\quad d\Lambda d(e^{236}) = e^{136} + (-1.0)e^{236} \quad B - F$

$C,\quad d\Lambda d(e^{145}) = e^{145} + (-1.0)e^{245} \quad C - G$

$G,\quad d\Lambda d(e^{245}) = e^{145} + (-1.0)e^{245} \quad C - G$

$D,\quad d\Lambda d(e^{146}) = (-1.0)e^{146} + (-1.0)e^{246} \quad -D - H$

$H, \quad d\Lambda d(e^{246}) = e^{146} + e^{246} \quad D + H$

ODE system

$\partial_t C = 0$

$\partial_t G = 0$

$\partial_t D = 0$

$\partial_t A = 0$

$\partial_t H = 0$

$\partial_t E = 0$

$\partial_t B = 0$

$\partial_t F = 0$

ODE system with closed 3-forms

$\partial_t C = 0$

$\partial_t G = 0$

$\partial_t D = 0$

$\partial_t A = 0$

$\partial_t H = 0$

$\partial_t E = 0$

$\partial_t B = 0$

$\partial_t F = 0$